

GIRI KANNAN M P

Salem,Tamilnadu - 636111 | girikannan10p@gmail.com | linkedin.com/in/girikannan-m-p-99a1b0334/ | github.com/Girikannan2107 | leetcode.com/u/Girikannan10/

SUMMARY

AI & Data Science engineering student with hands-on experience in Computer Vision, NLP, and Agentic AI. Built production-oriented ML systems achieving 85%+ accuracy and 90% reduction in manual processing time. IIT Palakkad hackathon finalist and SIH 2025 shortlistee. Proficient in Python, PyTorch, TensorFlow, Scikit-learn, FastAPI, and Docker.

TECHNICAL SKILLS

- Languages/ Development:** Python, Java, SQL, HTML/CSS
- Data Science & Analytics:**Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Data Preprocessing, Feature Engineering
- AI / ML & Deep Learning:**Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), TensorFlow, TensorFlow Lite, PyTorch, YOLO, OpenCV, CNN, RNN, LSTM, Transfer Learning, Model Optimization
- Databases/ Data Warehousing :** MySQL, PostgreSQL, SQLite, Snowflake, ChromaDB, MongoDB
- Tools/ Frameworks:** Git, GitHub, Kafka, Pathway, Tableau, Power BI
- Agentic AI & LLM Frameworks:** LangChain, LangGraph, Multi-Agent AI, Agentic AI, RAG
- Cloud, DevOps & Deployment:**AWS, Docker, FastAPI, Flask, Streamlit
- IoT Industrial AI:** IoT Sensors, Digital Twins, TinyML, Predictive Maintenance

EDUCATION

Karpagam College of Engineering	Sep 2023 - Present
Bachelor of Technology - Artificial Intelligence and Data Science, CGPA:8.33	
Holy Children Matric Hr Sec School	Jun 2021 - Mar 2023
Higher Secondary, Percentage:92.8	

PROJECTS

AI-Powered Handwritten Production Plan Digitization System

- Built an end-to-end OCR solution leveraging Python, OpenCV, DocTR, TrOCR, and OpenPyXL to digitize industrial production planning forms with 85%+ field extraction accuracy.
- Automated PDF-to-Excel conversion, cutting manual processing time by 90% and enabling structured data extraction, validation, and reporting for production workflows.

Solar-Hybrid Dewatering System with AI Digital Twin (SIH 2025)

- Designed a solar-hybrid mine dewatering solution using DC-coupled architecture, AI Digital Twin, and TinyML-based predictive maintenance to ensure near-zero downtime and reduce OPEX and diesel dependency.
- Enabled real-time monitoring and optimization using IoT sensors, weather data, and an OPEX (pay-per-use) model for ESG-compliant mining.

Regulatory-Grade Market Surveillance & Manipulation Detection System

- Developed a multi-layer surveillance framework using trade network analysis to detect manipulation patterns like circular trading and pump-and-dump.
- Built an explainable surveillance engine combining 20+ risk signals across trading behavior, timing patterns, and infrastructure usage to generate entity-level risk scores and investigation insights.

Aura Vision – Object Detection & Navigation Aid for the Blind

- Built an edge-based object detection system using YOLO, MobileNet-SSD, OpenCV, and TensorFlow Lite for real-time obstacle detection achieving 88% accuracy across 15+ object classes..
- Implemented low-latency audio guidance with real-time obstacle alerts, delivering inference speeds below 50 ms.

SERNOVA – AI & IoT-Driven Pregnancy Care Monitoring Platform

- Built an AI and IoT-based healthcare platform analyzing 12+ maternal and fetal health metrics, achieving 91% predictive accuracy for risk assessment and nutrition planning.
- Developed an agent-based decision system and interactive dashboard to enable continuous monitoring, adaptive care recommendations, and clinician-assisted interventions.

WORK EXPERIENCE

Python Intern - BYTS Services India Private Limited | Jun 2024 – Jul 2024

- Successfully completed an internship focused on data analysis and problem-solving using Python.
- Worked on real-world datasets to extract insights, improve data handling skills, and understand industry workflows.

ACHIEVEMENTS

Decodex 2026 National Level Hackathon, NLDIMSR, Second Place

Smartathon 2.0 L&T National Level Hackathon, Special Mention

Neurobots National level Hackathon, IIT Palakkad, Finalist

Smart India Hackathon 2025, Shortlisted

ADDITIONAL INFORMATION

Languages: English, Tamil

Certifications: Deep Learning, Data Analysis with Python - CognitiveClass.ai

Sports Achievement: State-level Kho-Kho Player – Represented at State Competitions